Yongyi Xie

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EDUCATION

Boston University College of Engineering **Boston, Massachusetts**
Master of Science in Electrical and Computer Engineering *Sep 2024 – May 2026*

Shanghai Jiao Tong University, UM-SJTU Joint Institute **Shanghai, China**
Bachelor of Engineering in Electrical and Computer Engineering *Sep 2020 – Aug 2024*

- Achieved Outstanding Student Award in Winter Program 2022
- Teaching Assistant for Discrete Mathematics and received recognition from Professor

SKILLS

- **Programming Languages:** Python, MATLAB, Java, Verilog, VHDL, C, C++, R
- **Data Analysis:** MySQL, Statistical analysis, data visualization, feature engineering
- **Machine Learning & AI:** PyTorch, Keras, TensorFlow, Model Evaluation, Hyper-parameter Fine Tuning, OpenAI

PROFESSIONAL EXPERIENCE

Amore AI Co. Ltd., Shenzhen, China ***Intern*** *May 2024 - Aug 2024*
Personal Companion AI with Customized Large Language Model
GitHub: https://github.com/Jn-Huang/ECE450CompanionLLMs

- Collaborated with fellows, designed a customized Chinese Companion AI for teenagers based on **ChatGLM** model.
- Implemented **Prompt Engineering** methods like **few-shot** and **chain-of-thought** reasoning to generate synthetic data, reaching same evaluation score as real-life data.
- Fine-tuned LLM models by **LoRA** using **llama-factory** in emotion understanding, reduced computation space effectively.

Fudan University, Shanghai, China ***Intern*** *Jan 2024 – Apr 2024*
Integrates Differential Gene Expression Analysis and Deep Learning for Prostate Cancer Diagnosis
GitHub: https://github.com/van1dh/Deep-Learning-Model-for-Prostate-Cancer-Diagnosis

- Directly report to the mentor, constructed a prostate cancer diagnosis model using **ANN**, increasing the diagnosis precision from 89% to 98%.
- Completed acquisition and pre-processing for Gene Expression Data using **R** for **Deep Learning Model** training, obtained 924 samples with 220 gene expression data each.
- Developed a deep learning model using **Python Keras** for prostate cancer diagnosis with 98% precision, outperforms traditional classifiers by 2%.
- **Publication:** Applied and Computational Engineering, vol. 57, no. 1, pp. 66–74, Apr. 2024

XiaoHua Semiconductor Co. Ltd., Shanghai, China ***Intern*** *Jul 2023 - Aug 2023*
RSA Encryption Circuit Design with Verilog

- Collaborated with MCU working team, focusing on microchip design using **Verilog** coding and testing methodologies.
- Designed and optimized an RSA encryption circuit using **Verilog**, reducing space and waiting time by 5%.
- Developed and analyzed sample code for **Finite State Machines**, and implemented testbench on a working portal.

EasyTravel: Travel Route Generation and Visualization App ***Project*** *Aug 2024 - Sep 2024*
GitHub: https://github.com/Yutong524/easytravel-front/tree/UC3

- Leading the front-end group, developed a travel planning application using **Google Maps API** to visualize travel routes.
- Implemented a database with **MySQL** to store and search **points of interest** (POIs) for personalized travel routes design.
- Developed algorithms to generate suitable routes based on visit order and user preferences.
- Created a clean, user-friendly interface utilizing **React** for seamless interaction.

MySpotify: Android Application Development with Kotlin ***Project*** *Nov 2023 - Dec 2023*

- Self-motivated, developed an music playing Android application, covering environment setup and **Kotlin** programming.
- Created and Implemented a mock **RESTFUL Api** json-server to handle the requests.
- Implemented the APP navigation using Jetpack Navigation to hold up the layout of the application.
- Developed an **MVVM**-structured user interface using **Jetpack Compose** to build the album/favorite UI.

FoodSearch: Online Food Ordering System based on Spring Boot ***Project*** *Sep 2023 - Oct 2023*

- Working on my own, developed an Online Food Ordering System using **Spring** and deployed the project via **AWS AppRunner**.
- Implemented **REST API** using **Spring MVC** for registration, menu search, ordering, and checkout.
- Used **Hibernate** for data access and storage for menus, restaurants information, etc.
- Built client-side with **ReactJS** and **Ant Design** for shopping cart functionality and order placement.